

SHABAA ALAM

Aeronautical Engineering Student | Imperial College London

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EDUCATION

Imperial College London — MEng Aeronautical Engineering (year 3 of 4)

Expected May 2028

- **Relevant Coursework:** Aerodynamics, Flight Dynamics, Structural Analysis, Control Systems, Advanced Manufacturing Processes.

TECHNICAL PROJECTS

ICLR Ember Rocket Team (EuRoC) / Design & Manufacturing Engineer

2025 – Present

- Designed full CAD assemblies for a custom filament winder in Fusion 360, adhering strictly to ISO tolerance standards.
- Precision-machined drive shaft components using manual lathes to achieve tight bearing tolerances within 5 microns.

Duocopter Flight Controller / 2nd Year Systems & Control Project

2026

- Modelled and simulated a discrete PID controller using MATLAB & Simulink to stabilize a dual-rotor unmanned aerial vehicle (UAV).
- Optimized control parameters to reduce response settling time and minimize total energy consumption during hovering operation.

Imperial Formula Racing (Formula Student) / Composites Engineer

2024 – 2025

- Engineered an ergonomic carbon fibre driver seat inlay using expanding foam, ensuring full compliance with strict competition safety regulations.
- Conducted risk assessments and structural safety checks prior to vehicle component manufacturing.

WORK EXPERIENCE & LEADERSHIP

Private & Academy Tutor (A-Level STEM) / Westcliff Academy for Learning

2023 – Present

- Communicate complex mathematical and physical concepts in A-Level Maths, Further Maths, and Physics to over 10 students weekly.
- Manage scheduling, client progress reporting, and customized lesson planning alongside a demanding full-time engineering degree.

Teaching Intern / Westcliff High School for Boys

Jun 2025 – Jul 2025

- Supported delivery of the secondary STEM curriculum and mentored students in structured, analytical problem-solving techniques.

TECHNICAL SKILLS & COMPETENCIES

Analysis & CFD/FEA:	Abaqus (FEA), STAR-CCM+, MATLAB, Simulink
CAD & Design:	SolidWorks, Fusion 360, Dassault 3DEXPERIENCE
Manufacturing:	Manual Lathes & Milling, Aluminium Casting, 3D Printing, Laser Cutting, Bandsaws
Languages:	English (Native), Bengali (Fluent)